# Prompt-Driven Visual Anomaly Detection for Retail Logistics via Off-the-Shelf VLMs


Junhui Wan[1] Hanyuan Chen[1] Fengliang Qi[1] Yuchen Hu[1] Qiuhui Shi[1] Tao Fang[1]

[1]Ant Group, Hangzhou, China

{hz_hanyuan.chy, bozhen.bb, wbz508256, huyuchen.hyc, qiuhui.sqh, xiaoyu.ft}@antgroup.com

*[Retail track]* Codabench: ant-hy



## Abstract

*We present an adaptive, prompt-driven approach for visual anomaly detection in retail logistics, targeting the VAND 3.0 Retail Track built on the Kaputt benchmark. Our method leverages off-the-shelf Vision-Language Models (VLMs) through a novel three-stage pipeline—Prompt Selector, Prompt Matcher, and Defect Analyzer—that dynamically selects and verifies domain-specific prompts conditioned on the material category of each query image. By encoding material-specific domain knowledge, structured chain-of-thought reasoning, and explicit judgment rules into the prompts, our approach achieves fine-grained defect analysis without any model training or parameter fine-tuning. Multiple VLMs produce independent defect scores, which are then fused via a Gaussian Naïve Bayes classifier with carefully engineered augmented features—including raw predictions, material encodings, statistical aggregates, rank dispersion, defect-rate interaction terms, and coarse-grained material indicators. Our method strictly qualifies as an Off-the-Shelf VLM submission under the competition rules, requiring no model training and only reference images at inference. Experimental results on the Kaputt benchmark demonstrate that our approach achieves competitive performance, validating the effectiveness of adaptive prompt engineering and ensemble-based score fusion for real-world retail defect detection. Code is available at* `https://github.com/Greywan/PD-VLM`.


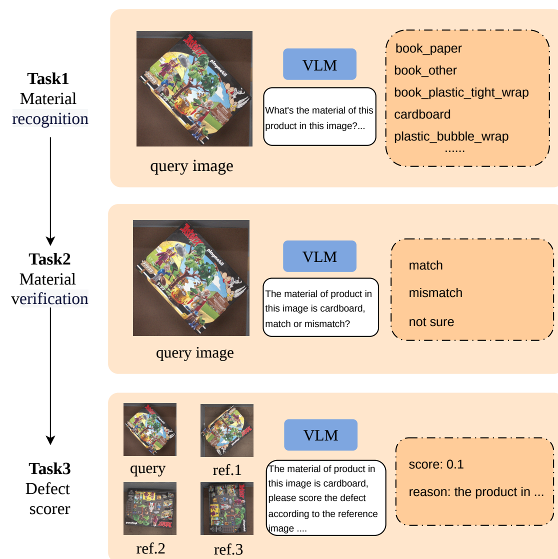


Figure 1. The proposed VLM-based framework for material-aware defect detection. The pipeline begins with material identification (Task 1) and consistency verification (Task 2) to provide a semantic prior. This leads to the core component, the Defect Scorer (Task 3), which performs a reference-based assessment. By comparing the query image with a set of reference samples, the VLM generates a quantitative defect score and a natural language explanation, enabling both precise and interpretable defect detection.

# 1. Introduction

## 1.1. Background

Traditional industrial defect detection benchmarks, such as MVTec-AD [8] and VisA [21], have primarily focused on controlled manufacturing scenarios characterized by fixed viewpoints, uniform illumination, and a limited number of object categories. State-of-the-art methods on these benchmarks have largely saturated, with AUROC scores commonly exceeding 99%. However, extending visual defect detection to large-scale retail logistics—such as warehousing and fulfillment centers—presents a fundamentally different and significantly more open-ended set of challenges.

In logistics centers, items are updated on a daily basis and defect samples are extremely rare in actual operations. This has driven the industry to rely heavily on more generalizable unsupervised anomaly detection (AD) methods or frontier Vision-Language Models (VLMs). Compared to manufacturing, anomaly detection in retail logistics faces new challenges, especially in terms of diversity and variability in object pose and appearance. Mainstream anomaly detection methods perform poorly when applied to this new scenario.

### 1.2. Challenge Description

This track requires participants to develop AI models capable of accurately identifying visual damage in real retail logistics applications, based on the **Kaputt** benchmark dataset [12], which comprises over 230,000 images and 48,000 unique products. The main challenges and difficulties of this competition include:

- **Extreme pose and structural variation:** Items are placed randomly on pallets, resulting in significant rotation, flipping, and other pose variations.
- **Complex material characteristics:** Packaging spans hard plastic, loose plastic bags, cardboard, books, and more. Some natural wrinkles (e.g., in plastic bags or book corners) or product designs (e.g., intentionally cracked book covers) are easily misidentified as anomalies by traditional models.
- **Fine-grained multi-class defects:** Defects include 7 types such as perforation, deformation, actuation, liquid or powder spillage, and missing multi-pack items. These are often extremely subtle and can be easily confused with lighting reflections.
- **Strict evaluation constraints:** The competition conducts blind evaluation on the fully undisclosed test set **Kaputt2**. The competition is divided into **Regular Setting** and **Off-the-Shelf VLM Setting** tracks, limiting large-scale systematic annotation, and uses **Average Precision ($\mathbf{AP_{any}}$)** as the sole ranking metric. This requires models to achieve extremely high sensitivity while controlling false positives (false alarms).

### 1.3. Contributions

Our contributions are as follows:

1. We propose a three-stage adaptive prompt selection and defect analysis pipeline that dynamically selects domain-specific prompts based on material classification, enabling fine-grained defect analysis with off-the-shelf VLMs.
2. We design a Gaussian Naïve Bayes ensemble method with engineered augmented features for fusing multi-VLM predictions, achieving robust defect probability estimation.
3. We demonstrate that our approach, which strictly qualifies as an Off-the-Shelf VLM submission (no training, with references), achieves competitive performance on the Kaputt benchmark for real-world retail defect detection.

## 2. Related Work

### 2.1. Traditional Anomaly Detection

Unsupervised anomaly detection methods have achieved remarkable success on industrial inspection benchmarks such as MVTec-AD [8] and VisA [21]. Feature-embedding approaches, exemplified by PatchCore [18] and its variants, construct memory banks of normal features and detect anomalies via nearest-neighbor distance, driving AUROC scores above 99% on these datasets. Reconstruction-based methods similarly excel in controlled settings. However, as demonstrated by the Kaputt dataset [12] and the VAND 3.0 challenge [11], these methods struggle significantly when transferred to the retail logistics domain, where the diversity of objects, poses, and packaging materials far exceeds that of traditional industrial benchmarks. The fundamental assumption of a stable set of normal prototypes breaks down in the face of thousands of ever-changing SKUs and extreme pose variation.

### 2.2. CLIP-based Zero-Shot Anomaly Detection

The advent of vision-language models, particularly CLIP [17], has catalyzed a new paradigm for zero-shot anomaly detection. **WinCLIP** [15] pioneered this direction by combining multi-scale sliding windows with text-visual alignment scores from CLIP, achieving zero-shot AUROC of 91.8% on MVTec-AD classification. While effective as a baseline, its purely contrastive approach lacks sensitivity to subtle defects. **AnomalyCLIP** [20] advanced the field by learning object-agnostic prompts on auxiliary data through a Discriminative Prompt Attention Module (DPAM), enabling better pixel-level localization and cross-category generalization—achieving 90.7% segmentation AUROC on MVTec-AD, a 24.9% improvement over WinCLIP. **AA-CLIP** [2] further refined this line by constructing anomaly-aware text anchors that explicitly model and separate normal vs. anomalous semantic spaces in a two-stage framework, outperforming AnomalyCLIP. **Bayesian Prompt Flow** [4] introduced probabilistic modeling of prompt uncertainty, learning prompt distributions rather than point estimates, which provides naturally calibrated probability outputs well-suited for ranking metrics like AP. **HEPE** [5] proposed hybrid explainable prompt enhancement for zero-shot industrial AD, incorporating interpretability into the prompt design process, which is particularly valuable for understanding model decisions in safety-critical logistics applications.

These CLIP-based methods share a common limitation in the retail setting: they were primarily designed for single-category industrial scenarios and lack the capacity to reason about the diverse material types and defect patterns encountered in logistics. Our approach addresses this by introducing material-adaptive prompt selection, which routes each image to domain-specific prompts tailored to its packaging characteristics.

### 2.3. VLM-based Anomaly Detection

The emergence of powerful VLMs has opened new possibilities for anomaly detection by leveraging their broad visual understanding and reasoning capabilities. **AnomalyGPT** [9] was the first method to employ large VLMs for industrial anomaly detection, producing interpretable dialog-style outputs that integrate anomaly localization and classification. While powerful, it requires supervised fine-tuning, making it incompatible with the Off-the-Shelf VLM setting. **Myriad** [1] proposed a modular architecture that plugs existing IAD methods as "visual experts" into a large multimodal model, where the LMM performs high-level reasoning and the experts provide precise detection—a design philosophy closely aligned with our multi-stage pipeline. **MMAD** [16] provides the first comprehensive benchmark for evaluating multimodal LLMs in industrial anomaly detection, revealing both the potential and current limitations of VLMs in this domain. **Fine-R1** [10] demonstrates that chain-of-thought reasoning can significantly enhance fine-grained visual recognition in multimodal LLMs, while Hong et al. [13] show that unlabeled data can improve fine-grained zero-shot classification.

More recently, **IAD-R1** [14] explored reinforcement learning to enforce consistent reasoning in industrial anomaly detection, and **AD-DINOv3** [19] enhanced DINOv3 with anomaly-aware calibration for zero-shot detection. **LogicAD** [6] introduced VLM-based text reasoning for explainable anomaly detection, leveraging logical inference chains to improve interpretability. **VELM** [7] proposed a multi-modal VLM pipeline specifically designed for anomaly classification (rather than just detection), which is relevant to the Kaputt benchmark's 7-type defect taxonomy.

Our work builds on these insights by incorporating structured chain-of-thought reasoning and domain knowledge into prompts, enabling off-the-shelf VLMs to perform precise defect analysis without any model training. Unlike methods that require fine-tuning (AnomalyGPT) or task-specific prompt learning (AnomalyCLIP, AA-CLIP), our approach achieves material-aware adaptation purely through dynamic prompt selection and multi-model ensemble.

### 2.4. Emerging Paradigms

Beyond the above categories, several emerging approaches deserve mention. Diffusion-based methods have been explored for training-free zero-shot anomaly localization, where a diffusion model generates "normal" reference images for comparison with test images. However, the fidelity of diffusion-generated references may be insufficient for the complex retail logistics scenarios in Kaputt. **AnomalyLVM** [3] proposed a vision-language model specifically designed for zero-shot anomaly detection, though its generalization to the extreme diversity of retail products remains an open question.

## 3. Methodology

### 3.1. Approach

Our method follows a two-level architecture: (1) a primary VLM undergoes a three-stage processing pipeline to produce a defect prediction score, and (2) additional VLMs directly produce independent defect scores, which are then fused via model ensemble.

#### 3.1.1. Three-Stage Processing Pipeline

**Stage 1 — Prompt Selector ($\mathcal{F}_{\text{sel}}$).** Classifies the input query image to obtain a material code label, determining which domain-specific prompt set to apply.

**Stage 2 — Prompt Matcher ($\mathcal{F}_{\text{mat}}$).** Verifies whether the selected material code's defect analysis prompt is appropriate for the current query image. If the match is insufficient (match score $< 0.7$), a fallback material code is selected instead.

**Stage 3 — Defect Analyzer ($\mathcal{F}_{\text{ana}}$).** Loads the corresponding analysis prompt based on the resolved material code, combines it with an output format prompt and reference images, and produces the final defect analysis result.

Through analysis of the dataset characteristics, we found that book-category items do not require fine-grained subcategories. Therefore, we merge `book_paper`, `book_plastic_tight_wrap`, and `book_other` into a single unified `book` category.

The overall three-stage procedure is summarized in Algorithm 1.

The key symbols used throughout this paper are defined in Table 1.

### 3.2. Prompt Design

#### 3.2.1. Stage 1 — Prompt Selector

The Prompt Selector receives a query image and classifies it into one of 8 material categories: `paper`, `cardboard`, `book`, `plastic_loose_bag`, `plastic_tight_wrap`, `plastic_bubble_wrap`, `plastic_hard`, and `other`. It outputs a material code, a confidence score, and a brief reasoning in strict JSON format.

**Algorithm 1** Three-Stage Adaptive Prompt Selection and Defect Analysis

**Require:** Query image $\mathbf{I}_q$, reference image $\mathbf{I}_r$
**Ensure:** Defect score $s$, explanation $\mathbf{e}$

1: **Stage 1: Prompt Selector**
2: $(m, c) \leftarrow \mathcal{F}_{\text{sel}}(\mathbf{I}_q)$ {Select material code $m$ with confidence $c$}
3: **Stage 2: Prompt Matcher**
4: $(\delta, \hat{m}) \leftarrow \mathcal{F}_{\text{mat}}(\mathbf{I}_q, m)$ {Verify match; fallback code $\hat{m}$}
5: $m^* \leftarrow \begin{cases} m & \text{if } \delta = \texttt{True} \\ \hat{m} & \text{otherwise} \end{cases}$ {Resolve final material code}
6: **Stage 3: Defect Analyzer**
7: $(s, \mathbf{e}) \leftarrow \mathcal{F}_{\text{ana}}(\mathbf{I}_q, \mathbf{I}_r, m^*)$ {Analyze defect with matched prompt}
8: **return** $(s, \mathbf{e})$

Table 1. Symbol definitions.

| Symbol | Meaning |
|---|---|
| $\mathbf{I}_q$ | Query image |
| $\mathbf{I}_r$ | Reference image |
| $\mathcal{F}_{\text{sel}}$ | Stage 1 selector (Prompt Selector) |
| $\mathcal{F}_{\text{mat}}$ | Stage 2 matcher (Prompt Matcher) |
| $\mathcal{F}_{\text{ana}}$ | Stage 3 analyzer (Defect Analyzer) |
| $m^*$ | Final resolved material code |
| $\delta$ | Match indicator |
| $\hat{m}$ | Fallback material code |

#### 3.2.2. Stage 2 — Prompt Matcher

The Prompt Matcher takes the query image and the initially selected material code as input, evaluating the match across three dimensions: (1) material appearance consistency, (2) packaging type match, and (3) defect analysis applicability. If the match score falls below 0.7, it provides a fallback material code.

#### 3.2.3. Stage 3 — Defect Analyzer

Each material-specific defect analysis prompt comprises three essential components:

- **Domain Knowledge:** Material-specific common defect patterns and visual characteristics.
- **Chain-of-Thought (CoT) Reasoning:** A structured checklist guiding the VLM through systematic visual inspection steps (e.g., corner flatness, surface integrity, spine condition, and reference comparison for books).
- **Judgment Rules:** Explicit scoring criteria mapping observed defects to numerical defect scores, preventing over- or under-estimation.

### 3.3. Gaussian Naïve Bayes Ensemble

Given $K$ base VLMs' prediction scores for each sample, we construct augmented feature vectors and train a Gaussian Naïve Bayes (GNB) classifier to output a fused defect probability. The $K$ raw predictions are expanded into a $2K+12$ dimensional feature vector, including:

1. **Raw predictions:** The $K$ original defect scores from each VLM;
2. **Material integer encoding:** The resolved material code encoded as an integer;
3. **Five statistics:** Mean, standard deviation, maximum, minimum, and range of the $K$ predictions;
4. **Rank dispersion:** Standard deviation and range of the rank order across predictions;
5. **Defect-rate interaction terms:** $K$ prediction $\times$ defect-rate interaction terms ($s_k \cdot d_m$), where $d_m$ is the empirical defect rate for the corresponding material in the training set;
6. **Four coarse-grained material binary indicators:** Binary variables indicating the broad material category.

Gaussian Naïve Bayes assumes that each feature is conditionally independent given the class label and follows a Gaussian distribution. During training, for each class and each feature dimension, the mean and variance are estimated on the validation set. At inference, the posterior probability $P(Y{=}1 \mid \mathbf{X})$ is computed to produce the final fused prediction probability.

The complete feature vector construction and GNB fusion algorithm are detailed in Algorithm 2 in the supplementary material.

### 3.4. System Architecture

Our overall system architecture consists of the following components:

1. **Material Classification Module:** A VLM-based material classifier that maps the query image to one of 8 material categories.
2. **Prompt Consistency Verification Module:** A second VLM call that verifies the appropriateness of the selected material prompt and provides a fallback mechanism when the match score is below threshold (0.7).
3. **Defect Analysis Module:** A VLM that receives the query image, reference image, and material-specific prompt (with domain knowledge, CoT reasoning, and judgment rules) to produce a defect score and explanation.
4. **Multi-VLM Scoring:** Multiple off-the-shelf VLMs independently produce defect scores for each sample, serving as the base predictions.
5. **GNB Score Fusion:** The augmented feature vector (constructed from the $K$ VLM predictions and metadata) is fed into the trained Gaussian Naïve Bayes classifier, which outputs the final fused defect probability.

The system requires no model training or parameter fine-tuning for the VLM components. Only the lightweight GNB classifier is trained on the validation set, preserving the Off-the-Shelf VLM eligibility.

## 3.5. Dataset & Evaluation

### 3.5.1. Benchmark Dataset Utilization

We use the Kaputt benchmark dataset [12] as provided by the challenge, including the reference images for each product SKU. Reference images are used at their original resolution without resizing or cropping, directly input into the VLMs alongside query images for comparison. No data augmentation is applied to the query or reference images. We perform material category consolidation: `book_paper`, `book_plastic_tight_wrap`, and `book_other` are merged into the unified `book` category based on empirical analysis showing that book subtypes do not benefit from fine-grained distinction.

### 3.5.2. Evaluation Criteria

The model's performance is evaluated using the following metrics:

- **Average Precision (AP)** — Primary ranking metric. Area under the precision-recall curve.
- **AUROC** — Area under the ROC curve. Provides an overall measure of discriminative ability.
- **Recall @ 50% Precision** — Recall when precision is at least 50%.
- **Recall @ 1% FPR** — Recall at 1% false positive rate, critical in high-throughput logistics settings.

### 3.5.3. Off-the-Shelf VLM Qualification

Our method qualifies as an Off-the-Shelf VLM submission because:

1. **No model training or fine-tuning:** All VLMs are used in their pre-trained, off-the-shelf form without any parameter updates.
2. **No task-specific model adaptation:** The only trainable component is the lightweight GNB classifier, which operates on the VLM output scores and metadata features—not on model parameters.
3. **Reference images only:** Our method uses only reference images provided by the benchmark at inference time, with no additional labeled training data for the VLMs themselves.

Our method corresponds to the **"No training, with references"** scenario as defined in the Kaputt paper [12]—we use few-shot anomaly detection with reference images, without any model training.

# 4. Experiments

## 4.1. Performance Metrics

Table 2 reports the performance of our full pipeline on the Kaputt benchmark. (Results will be updated upon release of official test scores from the Codabench evaluation platform.) "full" refers to the use of multiple models for fusion.

## 4.2. Ablation Study

We compare the following variants on a test sample set of 500 images in Table 3.

**clip classification**. Leveraging the remarkable zero-shot capabilities of the CLIP model, we adopt a defect classification approach to determine the presence of anomalies in query images. Specifically, for each defect category (e.g., Penetration, Deformation, Actuation, Deconstruction, etc.), we utilize Large Language Models (LLMs) to generate comprehensive descriptions, which serve as inputs for the CLIP text encoder. During the inference process, each query image is encoded via the image encoder; the resulting representation is then compared against all textual embeddings to compute similarity. After normalization, we obtain the final probability distribution across various defect types.

**capRL-based method**. CapRL is capable of generating dense captions for images, demonstrating a strong sensitivity to fine-grained visual details. Based on this model, we compared two distinct approaches: (1) CapRL direct output, which prompts the model to directly assign a score to the image; and (2) CapRL+LLM classification, a two-stage strategy where CapRL first generates a detailed textual description, and a Large Language Model (LLM) then determines the specific score based on that description.
*Note: Detailed numerical results for the above comparison table will be filled in upon completion of all experiments.*

# 5. Discussion

## 5.1. Challenges & Solutions

**Data noise.** The Kaputt dataset contains inherently noisy labels and ambiguous defect annotations, especially for borderline cases where natural packaging features (wrinkles, design elements) are indistinguishable from actual defects. Our three-stage prompt mechanism with material-specific judgment rules helps mitigate this by encoding precise defect criteria for each material type.

**Token consumption.** Multi-stage VLM inference with detailed CoT prompts consumes significant tokens per sample. We addressed this by optimizing prompt lengths while preserving critical domain knowledge and reasoning structure, balancing analysis quality with cost efficiency.

Table 2. Main results on the Kaputt2 benchmark.

| Method | $AP_{any}$ | AUROC | R@50%P | R@1%FPR |
|---|---|---|---|---|
| Ours (Qwen3-plus) | 0.4161 | 0.6457 | 0.3298 | 0.0580 |
| Ours (Gemini3-flash-preview) | 0.5067 | 0.7296 | 0.5876 | 0.0858 |
| Ours (full + GNB) | 0.6293 | 0.7872 | 0.6462 | 0.1937 |

Table 3. Ablation study on 500 test images.

| Method | |
|---|---|
| CLIP classification | 0.3500 |
| capRL direct output | 0.3900 |
| capRL + LLM classification | 0.3400 |
| IAD-R1 (industrial AD) | 0.3430 |
| Ours (Qwen3-plus) | 0.3985 |
| Ours (Gemini3-flash-preview) | 0.4211 |
| Ours (full + GNB) | 0.5894 |

Table 4. Off-the-Shelf VLM qualification criteria.

| Criterion | Compliance |
|---|---|
| No model training or fine-tuning | ✓ |
| No parameter updates on benchmark data | ✓ |
| Uses only reference images at inference | ✓ |
| No additional labeled data for VLMs | ✓ |
| Reproducible with publicly available models | ✓ |

All VLMs are used off-the-shelf; only the GNB classifier is trained on validation scores.

**Inference latency.** Running multiple VLMs per sample introduces considerable latency. We employed parallel inference across VLMs where possible and optimized the prompt matching stage to quickly reject mismatched material codes, reducing unnecessary computation.

### 5.2. Model Robustness & Adaptability

Our approach demonstrates robustness to real-world variations through several design choices:

- The **Prompt Matcher** stage acts as a safety net, catching misclassifications from the material selector and providing fallback prompts, which ensures that even incorrectly classified items receive a reasonable defect analysis.
- **Material-specific domain knowledge and judgment rules** are explicitly encoded in the prompts, reducing the VLM's reliance on its internal (potentially incomplete) knowledge about retail logistics defects.
- The **GNB ensemble** provides robustness across VLMs by leveraging their complementary strengths, with the augmented features capturing both individual model signals and cross-model consensus.

### 5.3. Off-the-Shelf VLM Qualification

We believe our submission strictly qualifies for the Best Off-the-Shelf VLMs (Retail) prize. Table 4 summarizes our compliance with each criterion.

The GNB classifier is a lightweight, non-VLM component that only operates on the output scores—it does not modify any VLM parameters and is a standard ensemble technique. The competition rules permit post-processing of VLM outputs, and our score fusion falls within this scope.

### 5.4. Future Work

- **Data analysis:** More comprehensive data analysis and refined defect definitions to better capture the nuance of retail logistics anomalies.
- **Data preprocessing:** Improved preprocessing and alignment for book-category items, where pose variation and cover symmetry pose unique challenges.
- **Inference efficiency:** Improving VLM inference efficiency, potentially using smaller-parameter models for prompt optimization stages.
- **Self-learning capability:** Developing automatic prompt optimization mechanisms that can iteratively refine prompts based on feedback.
- **Reference image learning strategy:** Current VLMs have limited capability in reference image injection and multi-image understanding. Future work could explore dedicated reference-aware VLM architectures or attention mechanisms.

## 6. Conclusion

We presented an adaptive, prompt-driven approach for visual anomaly detection in retail logistics that achieves competitive performance on the Kaputt benchmark without any model training. Our three-stage pipeline—comprising a Prompt Selector, Prompt Matcher, and Defect Analyzer—effectively leverages material-specific domain knowledge, structured chain-of-thought reasoning, and explicit judgment rules to enable off-the-shelf VLMs to perform fine-grained defect analysis. The Gaussian Naïve Bayes ensemble with engineered augmented features further improves prediction robustness by fusing complementary signals from multiple VLMs. Our method demonstrates that careful prompt engineering and multi-model ensemble strategies can unlock strong anomaly detection capabilities from existing VLMs, even in the challenging and highly variable retail logistics domain. The implications extend beyond this competition: as VLMs continue to improve in visual reasoning and multi-image understanding, adaptive prompt-driven approaches represent a promising direction for deploying AI-powered quality inspection in real-world, large-scale logistics operations.

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# A. Reproduction Details

This section provides the complete specifications of all model configurations, prompts, and algorithmic details necessary to independently reproduce our results.

## A.1. Three-Stage Pipeline VLM Configuration

The primary three-stage pipeline (Prompt Selector $\rightarrow$ Prompt Matcher $\rightarrow$ Defect Analyzer) is powered by **Gemini 3 Flash** with the following inference setting:

Table 5. Primary VLM configuration.

| Parameter | Value |
|---|---|
| Model | Gemini 3 Flash |
| Temperature | (to be filled) |
| Top-p | (to be filled) |
| Max output tokens | (to be filled) |

## A.2. Auxiliary VLM Configurations

In addition to the primary three-stage pipeline, auxiliary VLMs directly produce independent defect scores for ensemble. Each auxiliary VLM uses a reference-image-based prompting strategy with a shared prompt structure but model-specific reasoning instructions.

### A.2.1. Auxiliary VLM — GLM-4.6B

Table 6. GLM-4.6B configuration.

| Parameter | Value |
|---|---|
| Model | GLM-4.6B |
| Temperature | 0.7 |

**Reference Image Prompt Structure.** The GLM-4.6B auxiliary VLM uses the following prompt template:

- **Step 1:** Present the query image with the label “”
- **Step 2:** Present the reference image with the label “”
- **Step 3:** A detailed user prompt instructing the VLM to act as a retail logistics quality inspector, enumerating 7 defect types (penetration, deformation, actuation, deconstruction, spillage, superficial, missing unit) and outputting a JSON-formatted defect score with explanation.

### A.2.2. Auxiliary VLM — Qwen3.6-27B

**Reference Image Prompt Structure.** The Qwen3.6-27B auxiliary VLM employs a more structured 6-step reasoning chain:

1. Identify product and material type, recalling common defect patterns.
2. Check packaging integrity (most critical step).

Table 7. Qwen3.6-27B configuration.

| Parameter | Value |
|---|---|
| Model | Qwen3.6-27B |
| Temperature | 0.5 |

3. Count items for multi-pack products.
4. Inspect structure and surface.
5. Compare with reference image.
6. Output JSON-formatted result, biasing toward higher scores when uncertain.

## A.3. Gaussian Naïve Bayes Fusion Algorithm

Algorithm 2 provides the detailed specification of the GNB fusion module.

Table 8 summarizes the augmented feature vector construction.

Table 8. Feature vector summary ($2K + 12$ dimensions).

| Feature Group | Description | Dim |
|---|---|---|
| Raw predictions | Original defect scores | $K$ |
| Material encoding | Integer-encoded material | 1 |
| Statistical aggregates | Mean, std, max, min, range | 5 |
| Rank dispersion | Std and range of ranks | 2 |
| Defect-rate interactions | $s_k \times d_m$ per model | $K$ |
| Material indicators | Binary flags (4 categories) | 4 |
| **Total** | | $\mathbf{2K + 12}$ |

**Algorithm 2** Gaussian Naïve Bayes Multi-Model Score Fusion

**Require:** $K$ base models $\{f_1, \ldots, f_K\}$; validation set $\mathcal{D}_{\text{val}} = \{(\mathbf{c}_j, y_j)\}_{j=1}^{N}$ with labels $y_j \in \{0, 1\}$; test set $\mathcal{C}_{\text{test}}$; training-set material defect rate mapping $d_m$; label encoder le

**Ensure:** Fused prediction probability $p_{\text{fusion}}(\mathbf{c})$ for each test sample

**Phase 1: Feature Engineering** (per sample $\mathbf{c}$ with material $m^*$ and scores $\{s_1, \ldots, s_K\}$):

  Raw predictions: $[s_1, \ldots, s_K]$

  Material encoding: $m_{\text{enc}} = \text{le}(m^*)$

  Statistics: $\mu, \sigma, s_{\max}, s_{\min}, s_{\text{range}}$ of $\{s_k\}$

  Rank dispersion: $\text{std}(\text{rank}(s_k))$, $\text{range}(\text{rank}(s_k))$

  Defect-rate interactions: $s_k \times d_{m^*}$ for each $k$

  Material indicators: 4 binary flags $\rightarrow \mathbf{x} \in \mathbb{R}^{2K+12}$

**Phase 2: GNB Training** (on $\mathcal{D}_{\text{val}}$):

**for** each class $y \in \{0, 1\}$ **do**

  $\pi_y = |\{j : y_j = y\}|/N$

  **for** each feature dimension $d$ **do**

    $\mu_{y,d}$ = mean of feature $d$ for class $y$

    $\sigma^2_{y,d}$ = variance of feature $d$ for class $y$

  **end for**

**end for**

**Phase 3: Inference** (on $\mathcal{C}_{\text{test}}$):

**for** each test sample $\mathbf{c}$ **do**

  Compute augmented feature vector $\mathbf{x}$

  **for** each class $y \in \{0, 1\}$ **do**

    $\log P(\mathbf{x} \mid y) = \sum_d \left[-\frac{1}{2}\log(2\pi\sigma^2_{y,d}) - \frac{(x_d - \mu_{y,d})^2}{2\sigma^2_{y,d}}\right]$

    $\log P(y \mid \mathbf{x}) = \log \pi_y + \log P(\mathbf{x} \mid y)$

  **end for**

  $p_{\text{fusion}}(\mathbf{c}) = \frac{\exp(\log P(Y=1|\mathbf{x}))}{\exp(\log P(Y=0|\mathbf{x})) + \exp(\log P(Y=1|\mathbf{x}))}$

**end for**